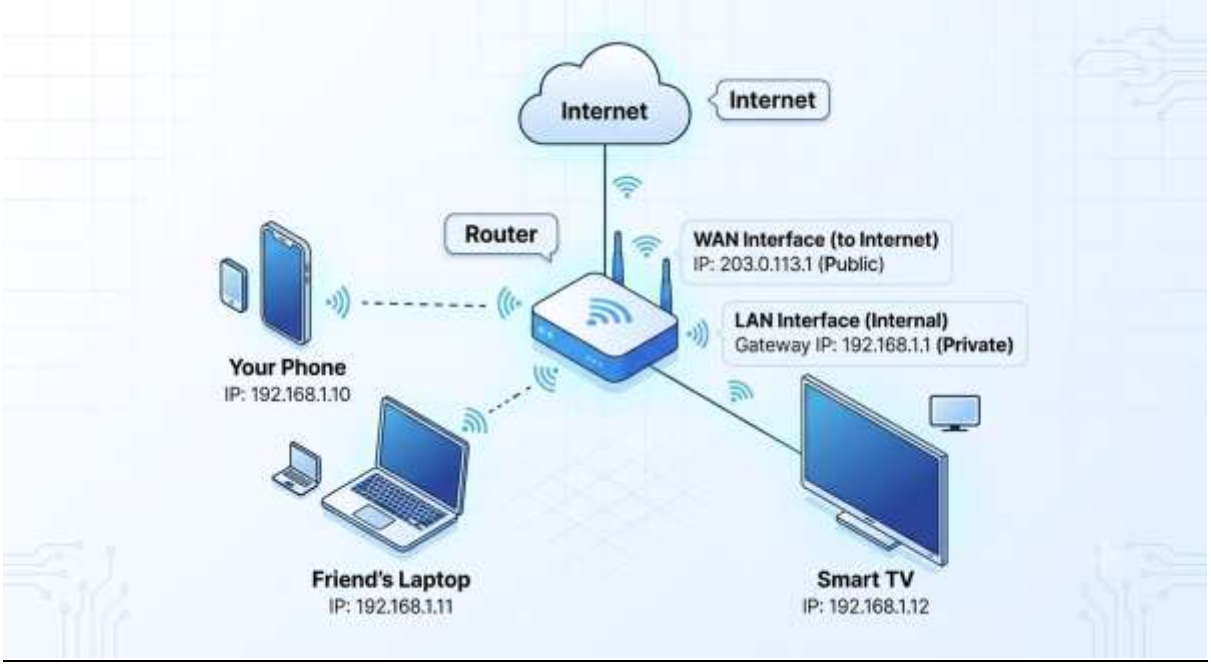
1.

Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.

ANS:-



2.

Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made. Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.

ANS:-

Simulated Manual IP Routing Between Different Subnets

Scenario Setup:

- **Source (Laptop):** IP 192.168.1.10, Subnet Mask 255.255.255.0, Default Gateway 192.168.1.1
- **Destination (Friend's Computer):** IP 10.0.5.25, Subnet Mask 255.255.255.0

Step-by-Step Packet Journey & Routing Decisions:

1. **Host Destination Check (Source Laptop):**
    - The laptop compares its own IP and subnet mask with the destination IP (10.0.5.25).
    - It performs a bitwise AND operation and recognizes that the destination is on a **different network subnet**.

- Because it is remote, the laptop knows it cannot communicate directly via ARP. Instead, it must forward the packet to its **Default Gateway** (Router A at 192.168.1.1).
- The laptop checks its ARP cache (or sends an ARP request) to find the MAC address of Router A, encapsulates the packet in an Ethernet frame, and transmits it.

2. **Hop 1: Router A (Local Gateway):**
   - **Routing Table Check at Router A:**
     - Destination 10.0.5.25 is matched against Router A's routing table.
     - Suppose the entry is: Network: 10.0.5.0/24 | Next Hop: 192.168.10.2 (Router B) | Interface: Serial0/0
   - **Routing Decision:** Router A strips the incoming Ethernet header, examines the network layer packet, and re-encapsulates it to forward across the WAN link toward the next-hop router (Router B).
3. **Hop 2: Router B (Destination Network Gateway):**
   - **Routing Table Check at Router B:**
     - Router B receives the packet on its WAN interface and checks its local routing table.
     - It finds a **directly connected route**: Network: 10.0.5.0/24 | Interface: GigabitEthernet0/1
   - **Routing Decision:** Since Router B is directly attached to the destination subnet, it uses ARP to resolve the MAC address of the friend's computer (10.0.5.25), delivers the frame over the local switch/network, and your friend's computer successfully receives the packet.

3.

Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

ANS:-

Routing Table for a Fictional Zomato-Style Delivery Network

Imagine Zomato operating three major regional branch offices (Mumbai, Ahmedabad, and Delhi) interconnected via routers:

Network Address	Subnet Mask	Next Hop	Interface
10.10.1.0	255.255.255.0	Direct (Connected)	GigabitEthernet0/0 (Mumbai Branch LAN)
10.10.2.0	255.255.255.0	10.10.1.254 (WAN Gateway)	Serial0/0 (Path to Ahmedabad Branch)
10.10.3.0	255.255.255.0	10.10.1.254 (WAN Gateway)	Serial0/0 (Path to Delhi Branch)
0.0.0.0 (Default)	0.0.0.0	192.168.100.1 (ISP Gateway)	GigabitEthernet0/1 (Internet Uplink)

4.

Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

ANS:-

Comparison: Static Routing vs. Dynamic Routing Protocols

Static Routing

- **Pros:**
    1. **High Security:** No routing update packets are exchanged over the network, making it immune to route poisoning or malicious spoofing attacks.
    2. **Predictable & Deterministic:** Paths are explicitly defined by the administrator, ensuring traffic always takes a predetermined route.
    3. **Zero Overhead:** Consumes minimal CPU and memory resources on routers since calculations aren't performed automatically.
- **Cons:**
    1. **Not Scalable:** Becomes extremely tedious and error-prone to manage manually in large networks with hundreds of subnets.
    2. **No Automatic Failover:** If a primary link fails, traffic drops permanently unless an administrator manually logs in and configures a backup route.
    3. **Rigid Maintenance:** Every time a new restaurant branch or server subnet is added to a food delivery network, static routes must be updated manually on every single router.

Dynamic Routing (e.g., OSPF, RIP)

- **Pros:**
    1. **Automatic Topology Adaptation:** Routers automatically discover network changes and calculate new paths if a link breaks (crucial for maintaining continuous order processing in apps like Zomato).
    2. **Scalability:** Effortlessly handles large, growing enterprise networks and multiple interconnected branches with thousands of nodes.
    3. **Load Balancing:** Advanced protocols can automatically split traffic across multiple equal-cost paths to optimize bandwidth.
- **Cons:**
    1. **Higher Resource Consumption:** Requires more router CPU, memory, and bandwidth to constantly exchange routing table updates (hello packets, link-state advertisements).
    2. **Configuration Complexity:** Setting up areas, process IDs, and authentication keys requires advanced networking expertise.
    3. **Vulnerability to Misconfiguration:** A routing loop or misconfigured metric in dynamic protocols can accidentally black-hole large portions of network traffic.

5.

Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does. Hint: Paste the AI-generated config and annotate each part with a short description.

ANS:-

OSPF Routing Configuration & Annotations

AI-Generated OSPF Sample Configuration (Cisco IOS):

```
router ospf 1
 router-id 1.1.1.1
 network 192.168.1.0 0.0.0.255 area 0
 network 10.10.10.0 0.0.0.3 area 0
 passive-interface GigabitEthernet0/1
```

Line-by-Line Annotations:

- router ospf 1: Enables the OSPF (Open Shortest Path First) routing process on the router with a process ID of 1 (locally significant).

- router-id 1.1.1.1: Assigns a unique 32-bit identifier to uniquely distinguish this router within the OSPF domain.
- network 192.168.1.0 0.0.0.255 area 0: Advertises the local LAN subnet (192.168.1.0 with wildcard mask 0.0.0.255) into OSPF Backbone **Area 0** so neighboring routers learn how to reach it.
- network 10.10.10.0 0.0.0.3 area 0: Enables OSPF on the point-to-point WAN link subnet using a /30 wildcard mask (0.0.0.3) within Area 0.
- passive-interface GigabitEthernet0/1: Prevents OSPF hello packets from being sent out of the specified interface (typically connected to end-user LANs), saving bandwidth and enhancing security.